

Curriculum Vitae

Basic Information:

- Name: Longtao Huangfu (皇甫龙韬)
- Gender: Male
- Date of Birth: December 1989
- Nationality: China
- Ethnicity: Han
- Political Affiliation: Communist Party Member
- Position: Assistant Professor / Research Associate Fellow/ PhD Supervisor,
Peking University Cancer Hospital / Gastrointestinal Surgery Center
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Education:

- September 2012 to July 2017, Harbin Medical University, Pharmacology, PhD
- September 2008 to July 2012, Harbin Medical University, Pharmacy, Bachelor

Work Experience:

- January 2025 to Present, Peking University, Peking University Cancer Hospital, Assistant Professor/ Research Associate Fellow
- August 2017 to December 2024, Peking University, Peking University Cancer Hospital, Research Assistant
- Present, Associate editor of Naunyn-Schmiedeberg's Archives of Pharmacology (Academic part-time job)

Research Interests:

The applicant is mentored by the renowned pharmacologist Academician Yang Baofeng. The current team is part of the Key Laboratory of Malignant Tumor Pathogenesis and Translational Research, led by Professor Ji Jiafu, and closely collaborates with the School of Pharmaceutical Sciences at Peking University to develop antitumor natural molecules and study tumor evolution. Core directions include:

1. Translational research of natural products guided by tumor evolution models for drug development strategies.
2. The clinical significance of extracellular DNA and its translational applications in anti-tumor therapy.
3. Research on anti-tumor combination therapies based on network pharmacology and quantitative systems pharmacology.

Addressing Bottlenecks:

Due to the limited variety of drugs for gastric cancer treatment, the ease of patient resistance, and the impact of side effects on health, the candidate has conducted exploratory research on key molecules in the gastric cancer pathogenic network, identifying potential targets such as the deubiquitinase USP15, the glycolytic rate-limiting enzyme HKDC1, and the serine/threonine protein kinase TOPK. Relevant projects have received funding from the National Natural Science Foundation and the Beijing Natural Science Foundation, with total research funding exceeding 1 million yuan. To date, the candidate has published 10 SCI papers as the first author and has been granted 4 patents, achieving significant breakthroughs and systematic results in the field of gastric cancer therapeutics.

Academic Positions:

- Member of the Tumor Marker Professional Committee of the Chinese Anti-Cancer Association, September 2018 - Present

- Member of the Abdominal Oncology Medicine Comprehensive Rehabilitation Branch of the Chinese Medical Education Association, July 2024 - Present

Research Achievements:

1. Funded Projects and Topics in the Last Five Years:

- i. National Natural Science Foundation, General Program, 82573870, "The Mechanism of Platinum-Induced Extrachromosomal DNA in Promoting Acquired Drug Resistance in Gastric Cancer and a Combination Therapy Strategy Targeting DNA-PK ", January 1, 2026 - December 31, 2029, 490,000 yuan (special) + 300,000 yuan (matching), ongoing, Principal Investigator.
- ii. National Natural Science Foundation, Youth Science Fund Project, 82203579, "Molecular Mechanisms and Intervention Strategies of Deubiquitinase USP15 Mediating Glucose Metabolism Promoting Gastric Cancer Development", January 1, 2023 - December 31, 2025, 300,000 yuan (special) + 300,000 yuan (matching), ongoing, Principal Investigator.
- iii. Peking University Cancer Hospital, Interdisciplinary Fund Project, JC202304, "Development of Antitumor Peptides Targeting Deubiquitinase USP15 and Its Pharmacological Mechanisms", January 2023 - December 2024, 300,000 yuan, ongoing, Principal Investigator.
- iv. Beijing Natural Science Foundation, Youth Science Fund Project, 7214215, "Mechanisms of Paclitaxel Activating Key Autophagy Molecule Beclin-1 Inhibiting Gastric Cancer Development and Combination Therapy Research", January 2021 - December 2022, 100,000 yuan, completed, Principal Investigator.

2. Major Participated Funded Projects and Topics in the Last Five Years:

- i. National Natural Science Foundation, General Project, 82272889, "Mechanisms and Intervention Strategies of SIRT1 Deacetylation Modifying Trim28/BRD4 Complex Reversing Chemotherapy Resistance in Gastric Cancer", January 1, 2023 - December 31, 2026, 520,000 yuan, ongoing, Participant.
- ii. National Natural Science Foundation, Joint Fund Project, U20A20371, "Multi-Omics Molecular Network Mapping and Mechanistic Exploration of Key Therapeutic Targets Based on Gastric Cancer Clinical Cohorts", January 1, 2021 - December 31, 2024, 2.6 million yuan, ongoing, Participant.
- iii. Beijing Natural Science Foundation, General Project, 7242020, "Molecular Mechanisms and Targeted Therapy Experimental Research of Gastric Tumor Microenvironment Fibroblast F13 Promoting Angiogenesis", January 1, 2024 - December 31, 2026, 200,000 yuan, ongoing, Participant.

3. Approved Patents:

- i. Huangfu Longtao; Xing Xiaofang; Guo Ting; Wang Yongqi; Ji Jiafu; "USP15 Protein HUBL Region Inhibitor of Compound EK143 and Its Application," August 23, 2024, China, ZL 2024 1 1167724.5.
- ii. Han Shuyan; Li Pingping; Hao Huifeng; Sun Hong; Xue Dong; Huangfu Longtao; "Peptide for Treating Tumors and Its Application," August 19, 2022, China, ZL 2022 1 0999027.0.
- iii. Huangfu Longtao; Xing Xiaofang; Ji Jiafu; "Filterable Microwave Reactor for Antitumor Drug Development," August 5, 2022, China, ZL202220851618.9.

- iv. Huangfu Longtao; Xing Xiaofang; Ji Jiafu; "Antitumor Drug Preparation Device," August 5, 2022, China, ZL202220895100.5.

4. Invited Academic Reports at Conferences:

- i. Longtao Huangfu; "Piceatannol Directly Targets Beclin-1 Promoting Apoptosis and Autophagy in Gastric Cancer", Korea International Gastric Cancer Week 2019, Incheon, Korea, April 11-13, 2019.
- ii. Longtao Huangfu; "Piceatannol Enhances Beclin-1 Activity to Suppress Tumor Progression and Its Combination Therapy Strategy with Everolimus in Gastric Cancer", XIV Sino-Russian International Pharmacology Conference, July 2023.
- iii. Longtao Huangfu; "The Mechanism of Cnicin Derivatives Targeting USP15 in Inhibiting Gastric Cancer and Organoid-Guided Drug Development", 97th Annual Meeting of the Japanese Gastric Cancer Association, Nagoya, Japan, March, 2025.

5. Published Articles:

- i. L. Huangfu*, H. Zhu*, G. Wang*, J. Chen*, Y. Wang, B. Fan, X. Wang, Q. Yao, T. Guo, J. Han, Y. Hu, H. Du, X. Li, J. Ji#, X. Xing#. "The Deubiquitinase USP15 Drives Malignant Progression of Gastric Cancer Through Glucose Metabolism Remodeling." J Exp Clin Cancer Res. August 20, 2024; 43(1):235. (IF: 11.4, 2024, Q1)
- ii. L. Huangfu*, X. Wang*, S. Tian, J. Chen, X. Wang, B. Fan, Q. Yao, G. Wang, C. Chen, J. Han, X. Xing# and J. Ji#. "Piceatannol Enhances Beclin-1 Activity to Suppress Tumor Progression and Its Combination Therapy Strategy with Everolimus in Gastric Cancer." SCI CHINA LIFE SCI 66 (2023), no. 2, 298-312. (IF: 8.0, 2023, Q1)

- iii. H. Zhu*, L. Huangfu*, J. Chen, J. Ji# and X. Xing#. "Exploring the Potential of Extrachromosomal DNA as a Novel Oncogenic Driver." *Science China Life Sciences*, 2025, 68(1): 144-157. (IF: 8.0, 2025, Q1)
- iv. S. Tian*, L. Huangfu*, S. Ai, J. Zheng, L. Shi, W. Yan, X. Zhu, Q. Wang, J. Deng, Y. Bao, S. Chang and L. Lu. "Causal Relationships Between Chronotype and Risk of Multiple Cancers by Using Longitudinal Data and Mendelian Randomization Analysis." *SCI CHINA LIFE SCI* 66 (2023), no. 10, 2433-2436. (IF: 8.0, 2023, Q1)
- v. Q. He*, L. Huangfu*, B. Fan*, Q. Zhuang, L. He, L. Li, W. You# and X. Xing#. "T-Cells Infiltration Mediates the Association Between Neutrophil/Lymphocyte Ratio and Survival in Gastric Cancer." *CANCER MED-US* 12 (2023), no. 15, 15893-15902. (IF: 2.9, 2023, Q2)
- vi. L. Huangfu*, B. Fan*, G. Wang, X. Gan, S. Tian, Q. He, Q. Yao, J. Shi, X. Li, H. Du, X. Gao, X. Xing# and J. Ji#. "Novel Prognostic Marker Linc00205 Promotes Tumorigenesis and Metastasis by Competitively Suppressing Mirna-26a in Gastric Cancer." *CELL DEATH DISCOVERY* 8 (2022), no. 51. (IF: 7.0, 2022, Q1)
- vii. G. Wang*, L. Huangfu*, X. Gao, X. Gan, X. Xing# and J. Ji#. "A Novel Classification and Scoring Method Based on Immune-Related Transcription Factor Regulation Patterns in Gastric Cancer." *FRONT ONCOL* 12 (2022), no. 887244. (IF: 4.7, 2022, Q2)
- viii. L. Huangfu, Q. He, J. Han, J. Shi, X. Li, X. Cheng, T. Guo, H. Du, W. Zhang, X. Gao, F. Luan, X. Xing# and J. Ji#. "MicroRNA-135b/CAMK2D Axis Contributes to Malignant Progression of Gastric Cancer Through EMT Process Remodeling." *INT J BIOL SCI* 17 (2021), no. 8, 1940-1952. (IF: 10.750, 2021, Q1)